AI-assisted accounting: controlled architecture comparison

DEVELOPMENT | INCOMPLETE

KNOWN STUDY DEFECTS / NOT FOR SELECTION

The original scores below are preserved for diagnosis. Complete data coverage does not make this run suitable for architecture selection. See the validity notice on the next page.

Run: pilot-v2 | Dataset: daybook-study-v1

Measured snapshot generated 2026-10-30T22:47:12.168Z. The PDF summarizes saved results only; exporting does not run models.

Scope	Measured state
Main episodes	101 recorded / 108 planned; 7 missing
Main split	development
Repeats per question	1
Verifier snapshot	development; complete; 80 planned candidates per method
Snapshot integrity	verified
Human readability score	Not evaluated. No total score out of 100 is computed.

How to read this report

This compares nine configurations of one accounting assistant. The ledger remains the source of financial numbers; retrieval supplies documentary context. Optional reviewers inspect candidates but do not replace deterministic accounting checks.

Structured task success requires the declared behavior, facts, scope, evidence and assumptions. It is a machine-checkable proxy, not proof that every sentence is accurate. The automated rubric has a maximum of 90; the remaining 10 points require human clarity review.

No winning architecture is selected here. Incomplete or development-only observations cannot establish a held-out improvement. Missing measurements remain N/A, and failed attempts remain in end-to-end denominators.

Known study defects - original results retained

Incomplete development pilot with a diagnosed transport race and pre-holdout control-protocol limitation

Notice recorded: 2026-10-30T22:48:54.206570+00:00

DAILY PROVIDER QUOTA

The run stopped at 101 of 108 planned episodes after four Gemini HTTP 429 daily free-tier quota responses and the persisted unknown-usage circuit opened. Seven episodes were not attempted. The provider identified a 500-request per-day, per-project, per-model quota. No retry replaced an outcome.

INFERRED COMPLETION READ RACE

R6 first cash episode reported RESEARCH_HTTP_FAILURE although its completed curl footer later showed HTTP 200 and exit 0. A non-atomic completion-file read is the supported explanation; the exact partial bytes were not captured. Original outcome and unknown-use reservation remain unchanged. A complete footer marker and partial-file regression tests are being added after this run.

CONTROL METADATA PROTOCOL

A pre-holdout code audit found R0 has no model-output fields or advertised vocabulary for required clarification/assumption tags. This is a structural limitation of the measurement adapter; the selected pilot does not establish how corrected clarification metadata will perform. The next development revision normalizes output metadata without adding financial tools, retrieval or a second model call.

This partial development record supports debugging and failure analysis, not architecture selection or a held-out accuracy claim. Source and full scoped ledger/document fingerprints were unchanged during the run. Missing attempts, provider failures and safe blocks stay visible; no winner is selected.

Original results SHA-256: 7db6a7ba258a4f3d377e1b220d63b2986c42097ca0916a9918c1a96253eb2ace

This exporter neither repairs prior answers nor rescales prior scores. A later development version must be reported separately. No improvement claim may be attributed to a defective comparison simply because its dataset remained unchanged.

Methods and experimental controls

Route	Query capability	Evidence retrieval	Answer review
R0	Fixed financial tools	No document retrieval	Code checks
R1	Broader analytical tools	No document retrieval	Code checks
R2	Same broader tools	Lexical documents	Code checks
R3	Same broader tools	Managed citation selection	Code checks
R4	Same broader tools	Embeddings + pgvector + lexical	Code checks
R5	Same broader tools	R4 + Jev reranking	Code checks
R6	Wren semantic SQL engine	Same custom hybrid	Code checks
R7	Same broader tools	Same custom hybrid	Code + LLM critic
R8	Same broader tools	Same custom hybrid	Code + Jev verifier

The shared synthetic ledger, document corpus, question groups and scoring reference are frozen. Candidate providers receive permitted questions and evidence, not hidden gold answers. Sources and queries are restricted to the authorized workspace; journal proposals are never automatically posted.

The study uses a common generator across routes; R0 reproduces the current tool architecture under that study model. R7 uses a separate critic context with the same model weights. Separate calls do not make generator and reviewer errors statistically independent.

R3 incurs an additional managed generation/search request. Its citations select authorized original documents; generated summaries are discarded. Opaque internal chunks and rankings prevent a clean algorithm-only retrieval comparison. R6 measures the restricted Wren engine integration, not the complete Wren product.

Failed requests are retained. Provider calls have bounded execution deadlines and no automatic retry. Pacing and local queue dispatch contribute to harness wall latency. Costs are paid-equivalent usage estimates plus reservations for unknown usage, not invoices or evidence that a free-tier request was charged.

Per-family outcomes and scenario-group bootstrap intervals preserve paraphrase/repeat clusters. Intervals are omitted when too few independent groups are available. Conditional reviewer metrics exclude provider failures; operational metrics block on failures and show that distinction.

Nine-route measured comparison

This development snapshot contains 101/108 planned episodes. Rows remain in predeclared route order. Provider errors and blocked answers remain in task-success denominators; missing episodes are disclosed rather than replaced.

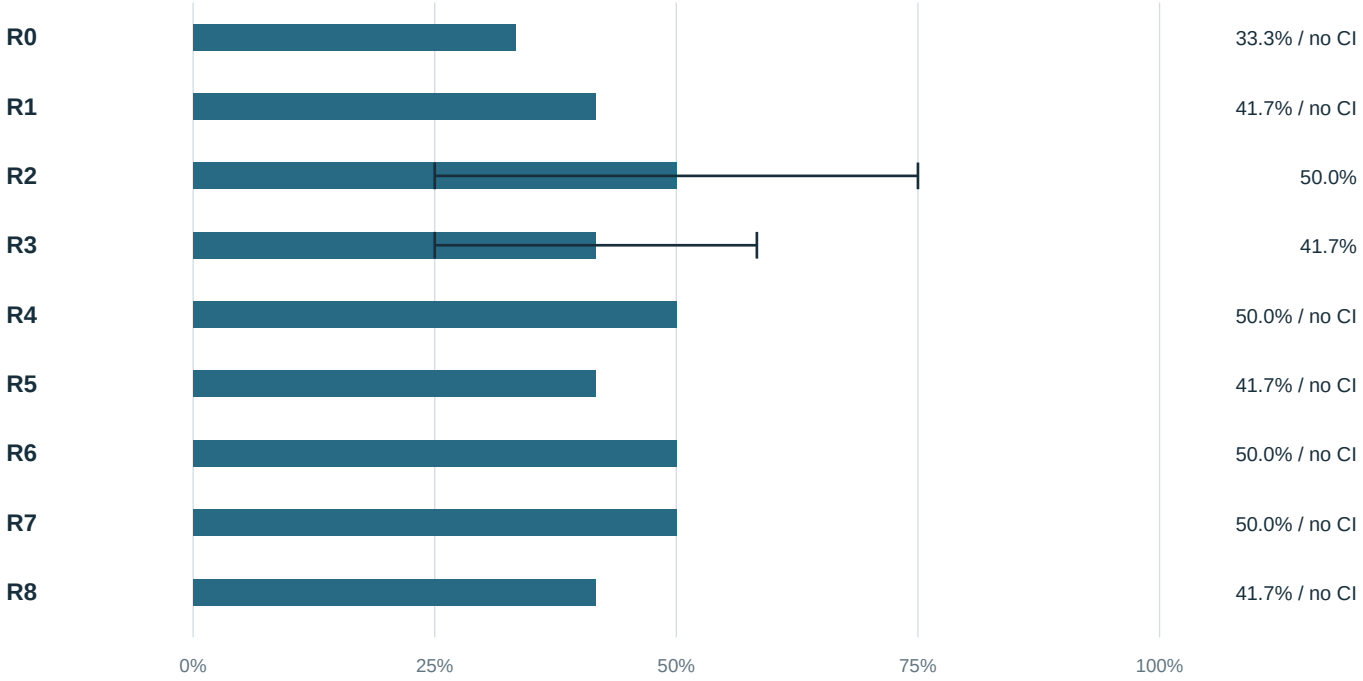
The all-family macro score gives equal weight to each of the six task families and stays N/A until every family has observations. The separate observed micro rate is useful for inspecting a partial run, but may change as later families are evaluated.

Route	n / planned	Micro success	All-family macro	Errors	Score / 90	Cost / episode
R0	11/12	36.4%	33.3%	27.3%	40.30	$0.0012
R1	11/12	45.5%	41.7%	0.0%	51.21	$0.0024
R2	12/12	50.0%	50.0%	8.3%	58.61	$0.0033
R3	12/12	41.7%	41.7%	16.7%	54.17	$0.0066
R4	11/12	54.5%	50.0%	0.0%	63.94	$0.0033
R5	11/12	45.5%	41.7%	0.0%	57.27	$0.0033
R6	11/12	54.5%	50.0%	18.2%	54.55	$0.0047
R7	11/12	54.5%	50.0%	0.0%	67.27	$0.0035
R8	11/12	45.5%	41.7%	9.1%	50.00	$0.0045

Micro success uses all recorded episodes. All-family macro is N/A until every task family is represented. Recorded-error rate includes transport/provider failures, not every blocked answer.

Task success across families

Figure 1. All-family task success and recorded 95% interval



Bars require all-family coverage. Whiskers reproduce the saved 95% scenario-group bootstrap interval; no interval is manufactured for a sparse pilot. N/A does not mean zero.

Figure 2. Structured task success within each task family

	financial	analytics	documents	scenario	entry	adversarial
R0	100% n=2	0% n=2	0% n=2	0% n=2	100% n=2	0% n=1
R1	100% n=2	50% n=2	0% n=2	0% n=2	100% n=2	0% n=1
R2	100% n=2	50% n=2	50% n=2	50% n=2	50% n=2	0% n=2
R3	50% n=2	0% n=2	50% n=2	50% n=2	100% n=2	0% n=2
R4	100% n=2	50% n=2	50% n=2	50% n=2	50% n=2	0% n=1
R5	100% n=2	0% n=2	50% n=2	50% n=2	50% n=2	0% n=1
R6	50% n=2	50% n=2	50% n=2	50% n=2	100% n=2	0% n=1
R7	100% n=2	0% n=2	50% n=2	50% n=2	100% n=2	0% n=1
R8	100% n=2	0% n=2	50% n=2	50% n=2	50% n=2	0% n=1

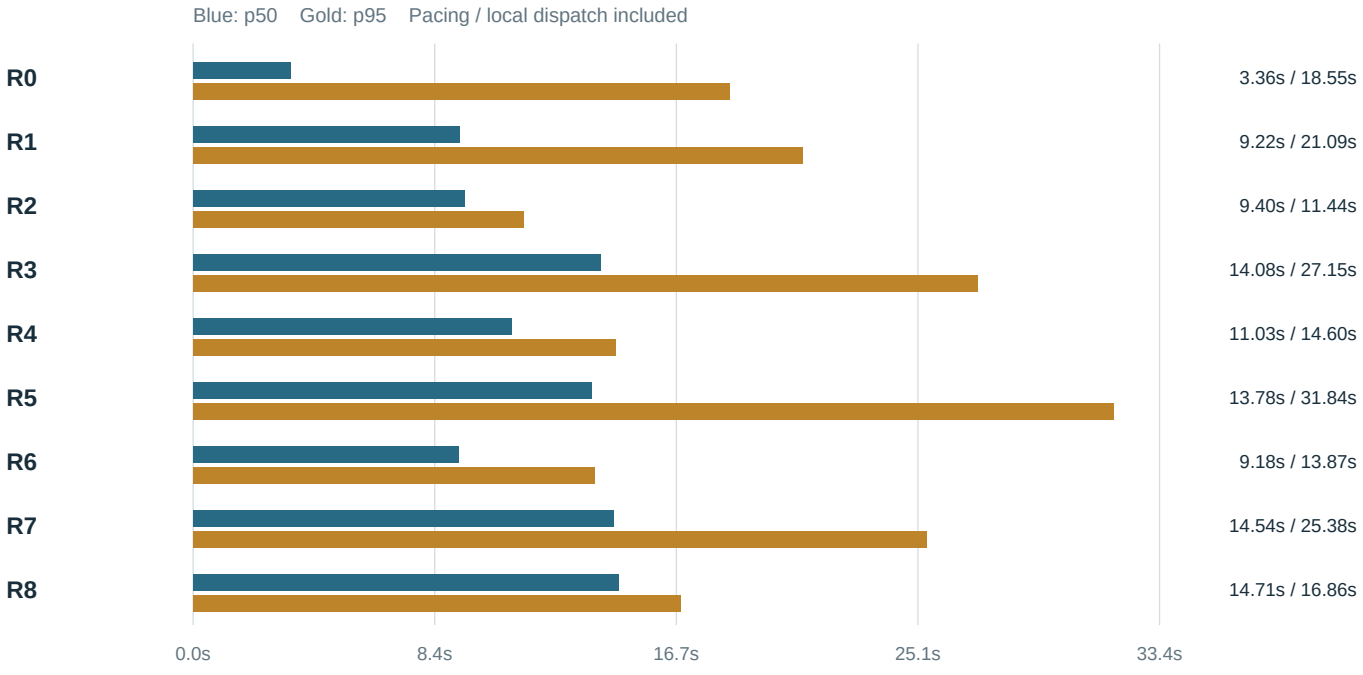
Each cell shows measured success and its episode denominator. Missing cells are N/A. Families have equal weight in the all-family macro score, despite possible unequal recorded counts during a partial run.

Cost and latency tradeoffs

Figure 3. Cost per recorded episode versus task success

The scatter uses observed micro success so partial data remain visible. It is descriptive, not a Pareto winner selection. Red marks indicate a recorded hard-gate disqualification. Missing cost or outcome values are omitted.

Figure 4. Harness wall latency: median and 95th percentile



Wall time includes provider calls, retrieval, validation, rate pacing and local dispatch delays. It does not establish production serving latency or million-user capacity.

Latency denominators and provider service time

Complete HTTP totals require measured service durations for every recorded provider attempt in an episode. Partial known subtotals are shown separately. Pacing and queue waits remain in wall time. These measurements are neither single-model latency nor production request latency. Non-error outcomes include blocks.

Route	Scope	n	Wall p50 / p95	Complete HTTP p50 / p95	HTTP C/P/U/N	Partial subtotal p50 / p95
R0	All	11	3.36s / 18.55s	0.81s / 16.62s	11/0/0/0	N/A / N/A
R0	Non-error	8	2.85s / 5.82s	0.77s / 2.15s	8/0/0/0	N/A / N/A
R1	All	11	9.22s / 21.09s	3.06s / 20.14s	11/0/0/0	N/A / N/A
R1	Non-error	11	9.22s / 21.09s	3.06s / 20.14s	11/0/0/0	N/A / N/A
R2	All	12	9.40s / 11.44s	3.12s / 4.64s	12/0/0/0	N/A / N/A
R2	Non-error	11	9.50s / 11.44s	3.19s / 4.64s	11/0/0/0	N/A / N/A
R3	All	12	14.08s / 27.15s	7.45s / 20.44s	11/0/0/1	N/A / N/A
R3	Non-error	10	14.08s / 18.46s	6.75s / 11.34s	10/0/0/0	N/A / N/A
R4	All	11	11.03s / 14.60s	3.34s / 5.80s	11/0/0/0	N/A / N/A
R4	Non-error	11	11.03s / 14.60s	3.34s / 5.80s	11/0/0/0	N/A / N/A
R5	All	11	13.78s / 31.84s	3.67s / 23.75s	11/0/0/0	N/A / N/A
R5	Non-error	11	13.78s / 31.84s	3.67s / 23.75s	11/0/0/0	N/A / N/A
R6	All	11	9.18s / 13.87s	2.45s / 4.68s	10/1/0/0	0.94s / 0.94s
R6	Non-error	9	9.69s / 13.87s	2.60s / 4.68s	9/0/0/0	N/A / N/A
R7	All	11	14.54s / 25.38s	3.84s / 13.90s	11/0/0/0	N/A / N/A
R7	Non-error	11	14.54s / 25.38s	3.84s / 13.90s	11/0/0/0	N/A / N/A
R8	All	11	14.71s / 16.86s	3.80s / 7.56s	11/0/0/0	N/A / N/A
R8	Non-error	10	14.71s / 16.86s	3.80s / 7.56s	10/0/0/0	N/A / N/A

C/P/U/N = complete / partial / unknown-or-unverified / no recorded calls. Complete-total percentiles exclude incomplete and no-call episodes. Partial-subtotal percentiles include only episodes with some measured calls and some missing timings. Missing durations never become zero. Legacy episode aggregates are unverified without per-call records; a timeout aggregate of 0 is not instantaneous HTTP execution.

Paired comparisons and cost accounting

Test	Comparison	Paired effect	Recorded 95% CI	Pairs	Unpaired T / C
H1	R1 - R0	8.3%	N/A	11	0 / 0
H2	R2 - R1	8.3%	N/A	11	1 / 0
H3a	R3 - R2	-8.3%	-25.0% to 8.3%	12	0 / 0
H3b	R3 - R4	-8.3%	N/A	11	1 / 0
H4	R4 - R2	0.0%	N/A	11	0 / 1
H5	R5 - R4	-8.3%	N/A	11	0 / 0
H6	R6 - R4	0.0%	N/A	11	0 / 0
H7a	R7 - R4	0.0%	N/A	11	0 / 0
H7b	R8 - R4	-8.3%	N/A	11	0 / 0

Effects are absolute percentage-point differences, macro-averaged over represented paired families. Positive favors the treatment. Sparse/incomplete comparisons do not cover every planned family; missing pairs and intervals are explicit. These multiple exploratory comparisons are not corrected into a single confirmatory significance claim.

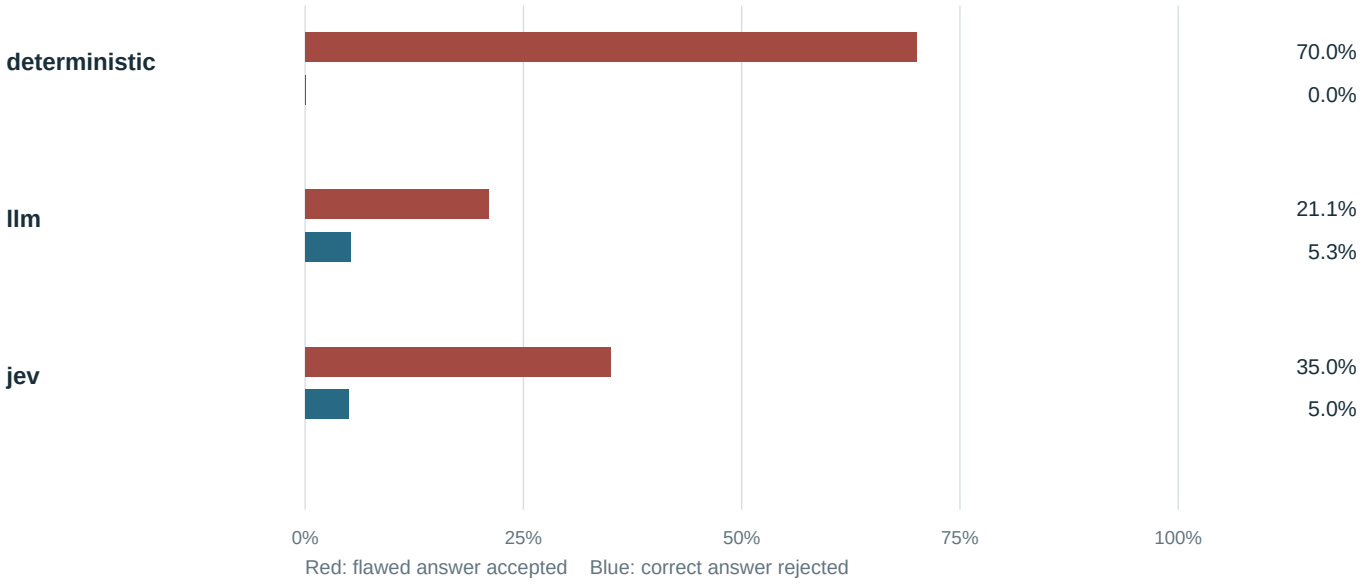
Route	Reported estimate	Unknown reservation	Unknown calls	Budget / success
R0	$0.0065	$0.0065	1	$0.0032
R1	$0.0261	$0.0000	0	$0.0052
R2	$0.0302	$0.0095	1	$0.0066
R3	$0.0441	$0.0346	1	$0.0157
R4	$0.0338	$0.0020	6	$0.0060
R5	$0.0337	$0.0023	7	$0.0072
R6	$0.0240	$0.0277	6	$0.0086
R7	$0.0371	$0.0017	5	$0.0065
R8	$0.0308	$0.0184	7	$0.0098

Setup/indexing budget: $0.0424. Unassigned run usage: $0.0000. A cost per success is N/A when there are no successes. Reservations retain uncertainty instead of declaring failed requests free.

Validator stress study - DEVELOPMENT

40 correct and 40 deliberately flawed candidate answers. Jev threshold: 0.8. Snapshot: COMPLETE.

Figure 5. Conditional reviewer errors on completed decisions



Reviewer	n / plan	Errors	Conditional FA / FR	Operational FA / FR	With code FA / FR
deterministic	80/80	0	70.0% / 0.0%	70.0% / 0.0%	70.0% / 0.0%
llm	80/80	4	21.1% / 5.3%	20.0% / 10.0%	20.0% / 10.0%
jev	80/80	0	35.0% / 5.0%	35.0% / 5.0%	27.5% / 5.0%

FA: flawed answer accepted. FR: correct answer rejected. Conditional denominators include completed reviewer judgments only. Operational metrics include provider failures as blocked outcomes; that is not proof that the reviewer detected the flaw. The combined column adds the deterministic checker.

Threshold selection uses development labels only, with a predeclared additional false-rejection constraint. Development results include threshold tuning and are not an unbiased held-out estimate. Maximum Jev hazard score is not a calibrated probability that the whole answer is wrong; no overall Brier score or human-judged truth score is invented.

These stress cases have 50% flawed prevalence and repeat source templates. They evaluate supplied candidate/evidence pairs, not production fraud prevalence, exhaustive search completeness, or independent human adjudication.

Supplemental retrieval and failure analysis

Conditional retrieval metrics include only document-applicable episodes with recorded completed retrieval. Unknown error stages and planner-skipped searches are counted separately; they are not silently scored as empty retrieval.

Route	Complete / eligible	Selected coverage	Recall@6	MRR@6	nDCG@6	E2E docs
R0	0/4	N/A	N/A	N/A	N/A	0.0%
R1	0/4	N/A	N/A	N/A	N/A	0.0%
R2	4/4	87.5%	87.5%	1.000	0.883	75.0%
R3	4/4	100.0%	N/A	N/A	N/A	75.0%
R4	4/4	100.0%	100.0%	1.000	0.943	75.0%
R5	4/4	100.0%	100.0%	1.000	0.980	75.0%
R6	4/4	100.0%	100.0%	1.000	0.943	100.0%
R7	4/4	87.5%	87.5%	1.000	0.883	75.0%
R8	4/4	100.0%	100.0%	1.000	0.963	75.0%

R3 managed citation selection has observable selected-document coverage; its raw search ranking metrics remain N/A. End-to-end document availability includes errors and blocks in the denominator, but does not imply answer correctness. Relevance uses exact required document/version IDs with binary gains.

Most frequent observed conditions

Condition	Episodes
structured_answer:required_fact_mismatch	26
structured_answer:missing_ledger_provenance	20
structured_answer:unsupported_or_unassessed_fact	20
structured_answer:required_fact_missing_or_multiple_mismatches	16
structured_answer:metric_key_mismatch	13
structured_answer:missing_document_evidence	13
structured_answer:wrong_behavior	13
structured_answer:missing_assumption	10
structured_answer:episode_error	9
capability:documents_unavailable	8
gate:deterministic_rejection	7
structured_answer:deterministic_validation_failed	7
structured_answer:missing_clarification_topic	6
operational:provider_rate_limit	4
structured_answer:proposal_line_mismatch	4

Conditions overlap and are not proven root causes. Key/schema/scope mismatch is not automatically an arithmetic error. Full per-episode records remain in the supplemental JSON.

Limitations and interpretation

1. All inputs are synthetic. References are independently implemented decimal fixture calculations, not audited business records. No independent human adjudication is claimed.

2. The primary score measures structured task success against frozen facts, scope, evidence IDs, proposal lines and explicit assumption/clarification tags. Natural-language summary faithfulness and usefulness still need human review.

3. Grounding scores measure source provenance: required source IDs, actual result-value bindings and deterministic calculations. They do not establish full semantic entailment of every citation. Supplemental claims without an independent reference or exact trusted-result binding are penalized as unassessed.

4. Automated scores are out of 90. The 10-point human clarity dimension and the combined 100-point score are unavailable; they are not imputed by another model.

5. Provider/tool errors count as failed end-to-end episodes. Conditional quality excludes status:error episodes and includes intentional blocks; it is not a replacement denominator.

6. Confidence intervals resample scenario groups within families. Paraphrases and repeat runs are correlated and stay together. Small synthetic samples do not establish real-world accuracy or safety.

7. All displayed 95% bootstrap intervals are unadjusted for the nine predeclared component contrasts and additional descriptive contrasts. They are exploratory effect estimates, not simultaneous confidence guarantees or claims of statistical significance. Two-group pilot and five-group-per-family held-out intervals may have poor coverage; zero-width empirical intervals do not establish certainty.

8. Reported monetary amounts are paid-equivalent estimates from returned usage and configured rates, not invoices. Unreported usage remains a separately identified conservative reservation; indexing/setup is separate.

9. Managed File Search uses the Interactions API to identify source citations. Its citation order is not a raw retrieval rank or exposed passage list. Authorized source IDs are mapped to the shared canonical documents before answer generation. This is managed source selection, not a like-for-like chunk retriever; raw-passage recall, MRR and nDCG are unavailable unless independently exposed.

10. Jev/LLM reviewers see supplied evidence, not ground truth, and can share generator errors. Their approval cannot override deterministic validation or database permissions.

11. The Wren route evaluates the pinned OSS semantic engine with the shared planner and approved views. It does not measure the complete commercial Wren product.

12. An incomplete route has no full-study conclusion. Missing episodes are reported separately, never silently removed or treated as free successful requests.

13. Episode completeness and snapshot integrity are separate gates. Changed or unverified before/after data makes this report ineligible for comparative selection even if every episode is present. Individual routes with hard-gate violations remain disqualified.

14. Queue-transport wall latency includes local dispatch delays and is not production request latency. Provider durations are measured separately.

15. All study generators normalized to Gemini3.5FlashLite; R0 is an architecture reproduction. Same-model critic errors may correlate.

16. Synthetic stress cases have 50% flawed prevalence; these results do not estimate production precision.

17. Correct/flawed pairs share source scenarios. Reused templates and fault patterns reduce independence; no publication-grade statistical power is claimed.

18. The deterministic baseline checks schema, citation existence and exact source amount/currency pairs. It intentionally cannot detect semantic period/entity/narrative contradictions.

19. Each semantic reviewer is run on every candidate, including code-detectable faults. Combined pipeline metrics show their additional contribution after code checks.

20. Provider errors are separate from semantic decisions. The operational column blocks on errors; the conditional column includes only completed reviewer responses.

21. Jev thresholds are selected only on development cases and frozen before holdout. Maximum hazard is an uncalibrated risk score; overall Brier calibration is unavailable.

22. LLM critic and main generator use the same Gemini3.5FlashLite weights in separate contexts; their errors are not statistically independent.

23. Reference fault labels are synthetic authored cases, not independently adjudicated human judgments. Narrative flaws are measured by these labels, not a judge-model truth claim.

Provenance and reproduction

All figures are vector graphics generated from the explicit saved JSON inputs. The exporter performs no provider requests, database queries, deployments or score recomputation. Source result values are retained; missing measurements remain N/A.

Exporter: daybook-pdf-report-v1. ReportLab: 5.1.0. Input bundle SHA-256: 3f4b00635504e2862b3f244c79b478cd93e7e8f322d73ba9a1e238ebbff759e1

main results: results.json

SHA-256: 7db6a7ba258a4f3d377e1b220d63b2986c42097ca0916a9918c1a96253eb2ace

verifier results: development-results.json

SHA-256: 141384fcc93b9e8dbd008b411b98ca419efeed4b501615a10181a11263b0da2f

validity notice: validity-notice.json

SHA-256: cb9a67afe93944a95cf9a43a6ffc94c7db1ac52af7ce0f22f57950bd6f053c6b

retrieval supplement: retrieval-and-failures.json

SHA-256: 46d3499414a02dac398452b817ecc3846a5c427a8b6107d35771521a11859f64

The accompanying provenance JSON binds these inputs, the exporter source, validation results and output PDF hash. Footer hashes identify this snapshot on every page.

Original result source revision: 07469a6ab8d2ca32b075bfb99ccbd264ddc05672. The working-tree/source manifest in the run directory identifies actual experimental code; a published ancestor alone is not sufficient.

Reproduce this export

```
.tools/report-render/bin/python scripts/export-research-report.py --main-results docs/evidence/research-runs/pilot-v2/results.json
--verifier-results docs/evidence/research-verifiers/verifiers-v1/development-results.json --verifier-split development --output
docs/evidence/research-runs/pilot-v2/study-report.pdf --supplement
docs/evidence/research-runs/pilot-v2/supplemental/retrieval-and-failures.json --validity-notice
docs/evidence/research-runs/pilot-v2/validity-notice.json
```

Renderer references: ReportLab Platypus and Graphics user guides (docs.reportlab.com/reportlab/userguide/ch5_platypus/ and ch11_graphics/). Scientific interpretation and experimental limitations come from the input study artifacts.